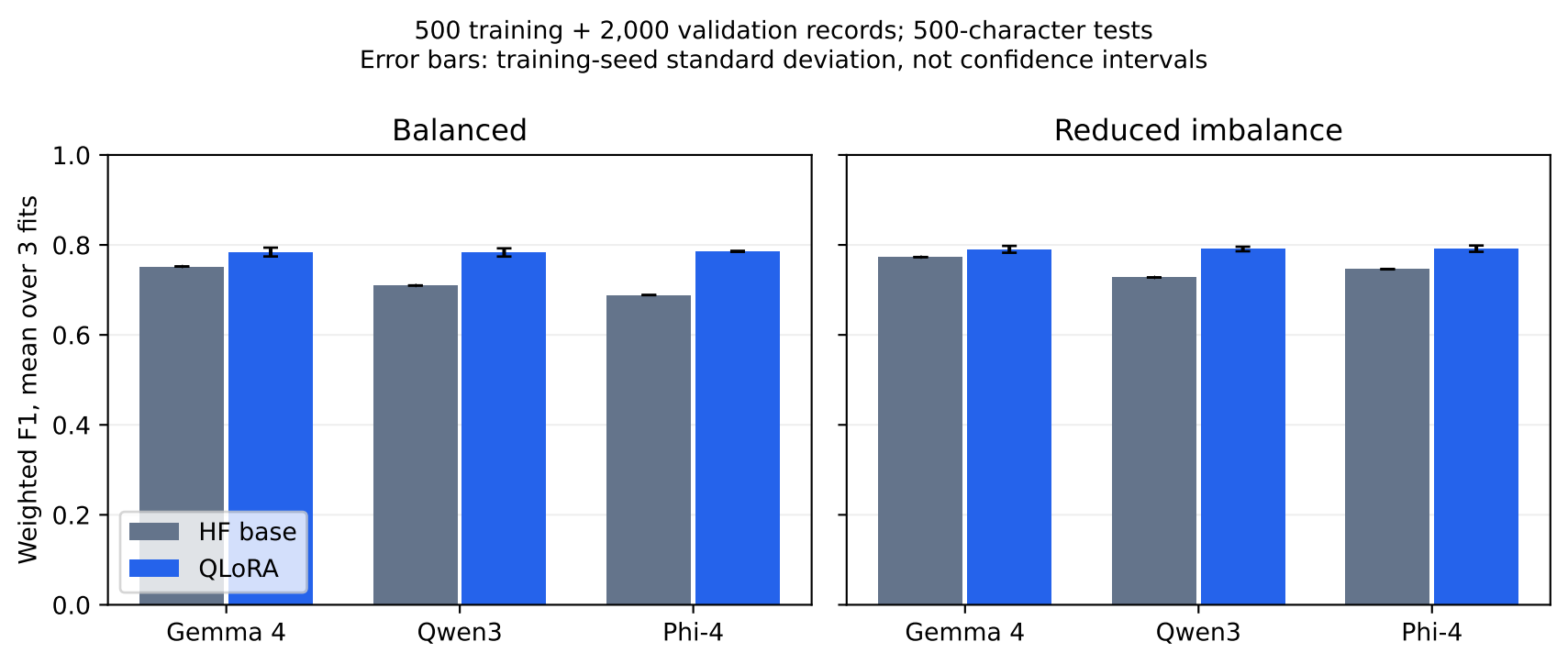
500 training + 2,000 validation records; 500-character tests
Error bars: training-seed standard deviation, not confidence intervals
Balanced
Reduced imbalance
Weighted F1, mean over 3 fits
1.0
0.8
0.6
0.4
0.2
0.0
HF base
QLoRA
Gemma 4
Qwen3
Phi-4
Gemma 4
Qwen3
Phi-4